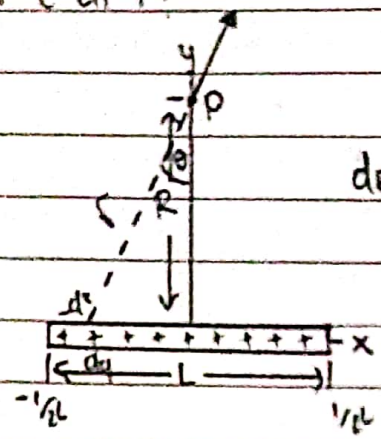
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1C - TNK

No.
Date

Tugas 4

Cari nilai E pada muatan garis rapat muatan garis λ.
Cari E di P?



$$dE = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r^2}$$

$$\Rightarrow E_p = \int dE\cos\theta = \int \frac{1}{4\pi\varepsilon_0}\frac{dQ}{r^2}\left(\frac{R}{r}\right)$$

$$\Rightarrow E_p = \frac{R}{4\pi\varepsilon_0}\int \frac{dq}{r^3}$$

$$\Rightarrow E_p = \frac{R}{4\pi\varepsilon_0}\int \frac{\lambda\,dx}{(x^2+R^2)^{3/2}}$$

$$\Rightarrow E_p = K\lambda R\int_{-\frac{1}{2}L}^{\frac{1}{2}L} \frac{dx}{(x^2+R^2)^{3/2}}$$

$$\Rightarrow E_p = K\lambda R\left[\frac{x}{R^2(x^2+R^2)^{3/2}}\right]_{-\frac{1}{2}L}^{\frac{1}{2}L} \Rightarrow E_p = \frac{K\lambda}{R}\left(\frac{L/2}{\sqrt{\frac{L^2}{4}+R^2}} - \frac{-L/2}{\sqrt{\frac{L^2}{4}+R^2}}\right)$$

$$\Rightarrow E_p = \frac{K\lambda}{R}\left(\frac{L/2}{\sqrt{\frac{L^2}{4}+R^2}} + \frac{L/2}{\sqrt{\frac{L^2}{4}+R^2}}\right)$$

$$\Rightarrow E_p = \frac{\lambda L}{4\pi\varepsilon_0 R\sqrt{\frac{L^2}{4}+R^2}} \rightarrow \vec{E} = \frac{\lambda L}{4\pi\varepsilon_0 R\sqrt{\frac{L^2}{4}+R^2}}\hat{j}$$

Dipindai dengan CamScanner